Software Engineering Assessment

Timber Inventory, Processing & Invoicing Management System

Objective: Design, build and develop a complete Timber Inventory, Processing, Stock Management and Invoicing Application. The assessment evaluates both coding ability and engineering thinking, including architecture, data integrity, concurrency, testing and traceability.

Complete lifecycle: Material Purchase → Material Addition → Processing → Stock Management → Product Sale → Invoice → Audit & Traceability

1. Business Scenario

A timber company purchases raw timber from suppliers. Raw timber is received and added to inventory, then processed into different finished products. Processing can result in finished products, secondary products, by-products, and waste/scrap.

Finished products are stored in inventory and subsequently sold to customers. When a product is sold, an invoice should be generated and the appropriate inventory should automatically be updated.

Workflow: Supplier → Raw Material → Processing → Finished Product → Inventory → Customer → Invoice

2. Raw Material Management

The application should allow users to add and manage raw materials. Information may include material name, material/SKU code, category/type, supplier, quantity, unit of measurement, purchase rate, purchase date, and batch/lot number.

When material is received, the corresponding inventory should automatically increase.

Material	Quantity	Unit	Purchase Rate
Teak Timber	1,000	CFT	₹2,000/CFT
Pine Timber	500	CFT	₹900/CFT
Plywood	100	Sheets	₹1,200/Sheet

3. Supplier Management

The application should maintain supplier information such as supplier/company name, contact person, phone, email, address and GST number. Material purchases should be associated with the appropriate supplier.

4. Processing / Production

Example – Processing Batch PB-001

Input: 500 CFT Raw Teak Timber

Output: 400 CFT Finished Teak Planks; 60 CFT Smaller Timber Pieces; 40 CFT Waste/Scrap.

After processing, the application should deduct consumed raw material, add finished and secondary products to inventory, record waste/scrap, record the processing batch, and maintain the relevant inventory transactions.

Processing should not be allowed when sufficient input material is unavailable. The design should support different combinations of input and output materials.

5. Finished Product Management

Products should contain information such as product name, product/SKU code, category, unit, selling price, tax/GST percentage and available quantity.

Product	Available Stock	Selling Rate
Finished Teak Plank	400 CFT	₹3,200/CFT
Teak Beam	150 CFT	₹3,500/CFT
Pine Plank	300 CFT	₹1,500/CFT

6. Stock Management

Every stock movement should create an inventory transaction. Movements may include material purchase, material consumption, processing output, customer sale, damage, waste, stock adjustment and invoice reversal. Inventory should not become negative. Direct stock modification without an appropriate transaction should be avoided.

Date	Item	Reference	Stock In	Stock Out	Balance
01-Oct	Raw Teak	Purchase PO-001	1,000	–	1,000
02-Oct	Raw Teak	Processing PB-001	–	500	500
02-Oct	Teak Plank	Processing PB-001	400	–	400
03-Oct	Teak Plank	Invoice INV-001	–	100	300

7. Customer Management

The application should maintain customer/company name, contact person, phone, email, billing address, shipping address and GST number.

8. Sales & Invoice Management

Customers should be able to purchase one or more finished products. Invoice totals should be calculated automatically. When an invoice is confirmed, finished-product stock should be deducted, inventory transactions should be created, and the sale should be associated with the customer. Selling quantities greater than available stock must be prevented.

Example Invoice INV-0001 – Customer: ABC Furniture Pvt Ltd

Product	Quantity	Rate	Amount
Finished Teak Plank	50 CFT	₹3,200	₹1,60,000
Teak Beam	20 CFT	₹3,500	₹70,000

Subtotal: ₹2,30,000 GST: As applicable Grand Total: Automatically calculated

9. Invoice Generation

Generate a printable invoice containing company information, unique invoice number, invoice date, customer and GST details, products, quantity, unit, rate, tax, subtotal and grand total. Downloadable PDF generation is preferred.

10. Invoice Cancellation & Stock Reversal

The application should support cancellation of a previously confirmed invoice. It should correctly handle invoice status, inventory reversal, transaction history, duplicate-cancellation prevention, auditability, and failure/rollback handling. Historical information should remain traceable.

11. Concurrent Stock Management

The system must correctly handle multiple users modifying stock simultaneously. Example: available Finished Teak Plank stock is 100 CFT; User A requests 70 CFT and User B requests 60 CFT at nearly the same time. The system must guarantee that inventory cannot become -30 CFT. The technical approach for maintaining consistency should be documented.

12. Batch & Lot Traceability

Maintain traceability across Supplier → Raw Material Lot → Processing Batch → Finished Product → Customer Invoice. The system should support forward traceability from a supplier lot to affected customer invoices, and backward traceability from a customer invoice to the originating supplier lot. This should be supported by the data model rather than manual notes.

13. Inventory Audit Trail

An administrator should be able to select any material or product and understand exactly how its current stock was calculated. Each inventory transaction should reference the business operation that created it.

14. Dashboard

Provide useful business information such as raw-material inventory, finished-product inventory, low-stock items, recent purchases, recent processing batches, recent invoices, monthly sales, inventory value, waste percentage and cancelled invoices.

15. Technical Requirements

The candidate may choose an appropriate technology stack. The solution should demonstrate frontend development, backend/API development, database design, business logic, application architecture, validation, error handling, database transactions, data integrity, concurrency handling, auditability, maintainable code and automated testing. Persistent storage/database is required.

16. API Design

Design appropriate APIs for materials, suppliers, purchases, processing, products, inventory, inventory history, customers, invoices, invoice cancellation and batch/lot traceability. The exact API structure and architecture are intentionally left to the candidate.

17. Testing Requirements

Automated tests should cover important business scenarios, including:

- Adding raw-material stock
- Processing material
- Insufficient raw-material stock
- Finished-product creation
- Invoice calculation and confirmation
- Finished-stock deduction
- Insufficient finished stock
- Invoice cancellation
- Stock reversal
- Duplicate cancellation
- Concurrent stock operations
- Batch traceability
- Invalid input
- Transaction rollback/failure scenarios

18. Deliverables

Please submit:
1. Complete source code
2. Frontend application
3. Backend/API
4. Database schema and migrations
5. Sample/seed data
6. Automated tests
7. README
8. API documentation
9. Architecture/design explanation

The README should include setup and run instructions, test instructions, technology choices, architecture overview, important assumptions, significant engineering decisions, and the approaches used for inventory consistency, concurrency and batch traceability. Docker/Docker Compose support is preferred but not mandatory. A deployed version is an advantage but not mandatory.

19. Evaluation Criteria

Area	What is evaluated
Problem Understanding	Ability to translate business requirements into software.
Architecture & System Design	Quality, maintainability and extensibility of the overall design.
Database Design	Appropriate entities, relationships, constraints and transaction modelling.
Code Quality	Readability, maintainability, organization, naming and abstractions.
Business Logic	Correct implementation of purchasing, processing, stock management, sales and invoicing.
Data Integrity	Accurate inventory across normal operations, failures and reversals.
Concurrency Handling	Consistency when multiple users modify inventory simultaneously.
Traceability	Ability to track products throughout the complete supply and production chain.
API & Frontend Engineering	Clear APIs, usable UI, validation, state management and error handling.
Testing & Documentation	Coverage of critical rules and clear communication of assumptions and technical decisions.

Final Expected Workflow

Supplier → Raw Timber Purchase → Raw Material / Lot Inventory → Processing → Finished Timber Inventory → Customer Sale → Invoice → Stock Deduction → Invoice Cancellation / Reversal (where applicable) → Complete Inventory & Batch Traceability

Important: This assessment is not intended to be a simple CRUD application. The objective is to demonstrate the ability to design and build a reliable business application where inventory, processing and financial transactions remain accurate, consistent and traceable. Where requirements are not explicitly defined, reasonable engineering assumptions may be made and should be documented.